

Command words

What each word asks you to do, and what the examiners reward

Write full sentences and give each point its reason or detail. Answer the question on the paper, using its scenario.

Which words carry the marks

Every Paper 1 and Paper 3 question part from 2022 to March 2026: 925 parts in 56 papers. A part counts under the command word in its own wording.

Word	Cambridge's definition (syllabus p. 66)	Parts	Share of marks	
Describe	state the points of a topic/give characteristics and main features	409	38.4%	
Explain	set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence	176	17.5%	
Discuss	write about issue(s) or topic(s) in depth in a structured way	80	15.4%	
Evaluate	judge or calculate the quality, importance, amount, or value of something	25	4.8%	
Analyse	examine in detail to show meaning, identify elements and the relationship between them	19	3.4%	
Justify	support a case with evidence/argument	20	3.1%	
Complete	not in the syllabus list; an instruction, not a command word	17	3.0%	
Identify	name/select/recognise	28	2.6%	
Compare	identify/comment on similarities and/or differences	16	2.3%	
Draw	not in the syllabus list; an instruction, not a command word	9	1.2%	
Write	not in the syllabus list; an instruction, not a command word	7	0.9%	
State	express in clear terms	12	0.8%	
Give	not in the syllabus list; an instruction, not a command word	11	0.8%	
Contrast	identify/comment on differences	6	0.6%	

"Write everything in bullet points" is wrong

1. No Cambridge document for 9626 tells you to answer in bullet points. The Cambridge guide for students tells you to answer in sentences.

"Answer the questions in sentences. Never give 'one word' or one 'phrase' answers"

(Cambridge Learner Guide for 9626, p. 15).

2. Examiner reports give this advice in almost every exam series. Use full sentences for Describe and Explain answers as well as for the long questions.

"Candidates should also be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences"

(Examiner report, November 2025, Paper 9626/32, p. 16). *"Candidates should write full sentences and avoid bulleted short statements in their responses"* (Examiner report, June 2022, Paper 9626/31, p. 13).

3. Bullets usually leave out what earns the marks. A bullet is often a bare point with no reason or detail. Most of the marks are for the reason or detail.

"the use of bullet points rarely produces little more than simple points or short statements with no explanations or descriptions" (Examiner report, June 2018, Paper 9626/32, p. 17). *"Short statements or bulleted responses that do not expand on facts cannot be awarded the higher marks because these are not discussions"*

(Examiner report, November 2022, Paper 9626/31, p. 14).

4. For Discuss, Evaluate and Analyse, the shape of the answer matters too. Bulleted lists and tables of statements cannot get most of the marks. Separate lists of benefits and drawbacks have the same problem when each item is one word or a very short sentence.

"learners who produced bulleted lists or tables of statements were unable to access most of the marks"

(Examiner report, March 2025, Paper 9626/32, p. 10). *"This comment also applies to answers that have separate lists of benefits and drawbacks, but the lists consist of single word or very brief sentences"*

(Examiner report, June 2025, Paper 9626/32, p. 22).

5. The mark schemes once printed this rule too. From 2021 to 2023, almost every Paper 1 and Paper 3 mark scheme limited bulleted answers to Discuss, Evaluate or Analyse questions. On an 8-mark question, the limit was 6 marks, however good the points were. Shorter questions had lower limits, for example 3 of 4 marks.

"Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion"

(Mark scheme 9626/12, March 2022, p. 4). *"five marks if bullets/list of points Must have expansions to be a proper analysis"* (Mark scheme 9626/12, March 2021, p. 9).

6. In 2024, only the March and June Paper 3 mark schemes still printed the limit. From November 2024, the mark schemes leave it out. The 2025 examiner reports still say that bulleted lists and tables cannot get most of the marks.

"Max 6 marks if bullets/list of points" (Mark scheme 9626/31, June 2024, p. 8). *"learners who produced bulleted lists or tables of statements were unable to access most of the marks"*

(Examiner report, March 2025, Paper 9626/32, p. 10).

7. One Paper 1 question in November 2025 asked for the steps of planning something. The examiners accepted a list of steps, sometimes bulleted, or a description. Each step had to show a decision. Treat this as a rare exception. Everywhere else, the examiner reports ask for full sentences.

"In the first, candidates listed (sometimes by creating a bulleted list) steps that would be taken, whilst others described the steps that had been taken. Either method was acceptable"

(Examiner report, November 2025, Paper 9626/12, p. 7).

How the marks are given

1. You do not lose marks for grammar or spelling. Examiners give marks for correct points and do not take

marks away for errors. Your meaning must still be clear.

"marks are not deducted for errors" (Mark scheme 9626/12, June 2025, p. 2). "answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous" (Mark scheme 9626/12, June 2025, p. 2).

2. Short sentences are fine. Put one clear point in each sentence. The examiner can mark it more easily, and you can check it more easily.

"Keep most sentences short and focused. That will help with clarity and in making every point stand out for the examiner" (Cambridge Learner Guide for 9626, p. 15).

3. Some mark schemes give one mark for a point and a further mark only for its reason or detail. They label the first mark (1st), with the linked follow-on marks under it. Some mark schemes say that the follow-on marks need the first one. One examiner report says the further marks were 'dependant on this answer'. Cambridge's mark-scheme key does not define (1st).

"(only award if either of previous two marks given)" (Mark scheme 9626/12, November 2025, p. 6). "Two further marks were then dependant on this answer" (Examiner report, June 2024, Paper 9626/11, p. 4).

4. One mark scheme limits an answer made only of bare points. On that 3-mark question, answers that are all first marks get at most 2 marks.

"MAX two marks if answers are all only first marks" (Mark scheme 9626/13, November 2023, p. 4).

5. Answer the question that the paper asks, about the scenario it gives. The full range of marks is only for answers that use the scenario. Do not write everything you know about one word in the question.

"the full range of marks is only available to candidates for answers referring to the scenario in the questions" (Examiner report, November 2025, Paper 9626/32, p. 16). "Centres must ensure that candidates target their answers to the questions as they are written and not at 'key' words in the questions" (Examiner report, June 2024, Paper 9626/31, p. 17).

6. Do not write out mark scheme bullets that you learnt by heart. Examiners notice this. A list of learnt points does not answer a new question.

"Examiners saw many scripts where candidates listed sets of bullet points that appeared to have been learnt directly from previous mark schemes" (Examiner report, November 2019, Paper 9626/11, p. 1).

7. Use the number of marks as a clue to how much to write. It tells you how many separate points, or how much detail, the examiner expects.

"Look at how many marks are allocated to a question. This gives you a clue as to how much work is expected" (Cambridge Learner Guide for 9626, p. 15). "the number of marks available for each question is a useful indicator of how much detail or how many descriptions should be given in the answers" (Examiner report, March 2025, Paper 9626/32, p. 8).

8. Never leave a question blank. A few sentences on a topic you are unsure of can still earn marks.

"Writing a few sentences, even if little is known about the topic, may gain valuable marks towards the overall total" (Examiner report, November 2025, Paper 9626/32, p. 16).

The words at a glance

Command word	What to do
Describe	Say what it is, what it does or how it happens, with one separate point for each mark. Give reasons only when the question asks for them.
Explain	Make a point, then give its reason with a linking word such as 'because' or 'so'. Do this for each mark you need.
Discuss	Write full sentences in paragraphs. Give benefits and drawbacks of the thing in the question, and aim for at least two of each. Expand every point with 'because' or 'so'.
Evaluate (and Assess)	Give points for AND against. Add a consequence to each point and say how much it matters in this scenario. End with a reasoned overall judgement.
Analyse	Break the topic into its parts. For each part, say what it leads to in the situation in the question: point, then 'so', then the consequence.
Justify	Argue only FOR the choice in the question. State a benefit, then give the reason it helps in this scenario. Never list drawbacks of the choice.
Compare and Contrast	For Compare, give similarities and differences. For Contrast, give differences only, and write every point about both things.
Identify, State, Give, Define, Name, Suggest	Give the exact term or fact, in the number of items the question asks for. Leave out explanations unless the question also says describe or explain.
Complete, Draw, Write (algorithms and flowcharts)	Use the syllabus keywords and flowchart symbols. Set counters before the loop, update them inside it, and close every structure.

Describe

Describe. Cambridge's definition: "state the points of a topic/give characteristics and main features"

Syllabus 2025 to 2027, p. 66

Check what you must describe

1. Describe asks what something is, what it does or how it happens. Explain adds why. You need a reason in a Describe answer only when the question asks for one, for example 'Describe, with reasons'. Then you cannot get full marks without the reasons.
"Explain set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence" (Syllabus 2025 to 2027, p. 66). "Must have at least TWO reasons to obtain full marks" (Mark scheme 9626/11, June 2023, p. 5).
2. Examiner reports often say that students mix up Describe and Explain. The mix-up costs marks in both directions. If the question asks why, a description of the thing itself gets few or no marks.
"Many candidates failed to differentiate between, for example, questions where they were asked to describe and questions where they were asked to explain" (Examiner report, June 2024, Paper 9626/11, p. 1). "where a question asks why a protocol or process would be used, there is usually little or no credit given for describing the protocol or process itself since this does not answer the question 'why?'" (Examiner report, June 2022, Paper 9626/31, p. 13).
3. For 'Describe how X is set up' or 'the process of', give the steps in order. Details of how X is used earn nothing there.
"Good answers detailed the steps that would be taken to set up an electronic wallet. No details of the use of the wallet were required to gain marks" (Examiner report, June 2022, Paper 9626/32, p. 16).
4. For 'Describe the features' or 'the functions', say what X is or does. For 'Describe the benefits', give benefits and leave out uses. An answer about the wrong one gets no marks, even if it is true.
"some focused on the use of the internet and remote learning, neither of which are features of a MOOC" (Examiner report, November 2022, Paper 9626/31, p. 14). "Common mistakes included descriptions of uses, inaccurate references to network performance and to illegal activities" (Examiner report, March 2025, Paper 9626/32, p. 9).

One mark for each point

1. In most Describe mark schemes, each correct point earns one mark. A 4-mark Describe question needs four separate points. One long sentence with one point in it is not enough.
"As a description is generally a one-mark answer, candidates had to give three separate descriptions for full marks to be awarded" (Examiner report, November 2025, Paper 9626/11, p. 1). "Question 1 has four marks available for describing the structure of a cellular network so at least four points need to be made in the answer for full marks" (Examiner report, November 2025, Paper 9626/33, p. 19).
2. Some mark schemes give one mark for a point, labelled (1st), and another for more detail about it, labelled (1). The (1st) label works as in the general rules at the front. On those questions, four short points can get 4 marks. Two points with more detail can also get 4 marks. 83 of the 388 Describe mark schemes from 2022 onward use (1st). None before 2022 use it.
"It is interrogated by the inference engine (1st) – Using relevant rules/rules that apply (1)" (Mark scheme 9626/11, November 2025, p. 5). "This meant that candidates could gain full credit by either giving four descriptions or expanding on fewer descriptions" (Examiner report, June 2025, Paper 9626/33, p. 26).
3. If the question asks for a set number of things, for example 'Describe two reasons', the examiner marks only that many answers. Extra answers do not help. If you join two ideas with 'and', the second idea can count as a separate answer. Give each point its own sentence.
"If, for example, the question asks for two reasons why an action may be taken, only the first two answers given by the candidate may be awarded marks" (Examiner report, November 2025, Paper 9626/11, p. 1). "Typically, where candidates

join two concepts together using 'and', the concept after the word 'and' is treated as a second answer, unless the piece after the 'and' is clearly an explanation or clarification of the piece before the 'and'"

(Examiner report, November 2025, Paper 9626/11, p. 1).

Say what each thing is or does

1. Say what each thing is or does. Examiner reports say that answers which only name components do not get the higher marks. One mark scheme allows a maximum of 3 of 5 marks for naming events without descriptions.

"Max three for naming valid events without descriptions" (Mark scheme 9626/32, June 2024, p. 6). "Many responses indicated that candidates knew the names of the components but were unable to describe them in the detail required to gain the higher marks" (Examiner report, June 2022, Paper 9626/13, p. 9).

2. Do not repeat the question. Answers such as 'White box testing is used to test the code' or 'a sound sensor measures sound' get no mark. Add a detail that the question does not give you.

"Further, many candidates repeated the question, writing that 'white box testing is used to test the code'. In general, statements that repeat the question are not given credit" (Examiner report, June 2024, Paper 9626/32, p. 22).

"Candidates should avoid simply repeating the question in their response without further expansion; for example, a humidity sensor measures humidity and a sound sensor measures sound"

(Examiner report, June 2022, Paper 9626/12, p. 5).

3. Vague words such as 'more expensive' get no marks on their own. Name the exact cost, or say exactly what is cheaper.

"Some weaker responses contained responses such as 'more expensive' which is not enough at this level and statements such as this should always be expanded" (Examiner report, June 2022, Paper 9626/13, p. 8). "However, where candidates stated that a specific, named, cost could be avoided, this was awarded"

(Examiner report, June 2025, Paper 9626/12, p. 6).

4. Use the correct technical term. Examiners say that slang often makes an answer meaningless.

"The use of what may be considered slang often renders answers meaningless"

(Examiner report, March 2025, Paper 9626/12, p. 1).

5. Write descriptions as full sentences, with one point in each sentence. The Paper 3 reports say that a description needs full sentences. On one Paper 1 question, the examiners accepted a list of steps for a 'describe the process' question. Full sentences are the safe choice on both papers.

"Descriptions or explanations can only be conveyed in full sentences"

(Examiner report, June 2022, Paper 9626/31, p. 13). *"In the first, candidates listed (sometimes by creating a bulleted list) steps that would be taken, whilst others described the steps that had been taken. Either method was acceptable"*

(Examiner report, November 2025, Paper 9626/12, p. 7).

Use the scenario

1. Fit your points to the scenario. The question may be about a nuclear power plant or a bank's new customers. A point that is true in general but wrong for the scenario loses the mark. A prepared answer that ignores the question gets only low marks.

"context also provides a restriction, as certain answers may be acceptable for one scenario, but not for another"

(Examiner report, November 2025, Paper 9626/11, p. 1). *"those who provided an answer they had prepared for, rather*

Worked example

Q4(b) [4 marks] "(b) Describe how a knowledge base is used in an expert system."

Question paper 9626/11, November 2025, p. 4

Part (a) asked what is meant by the term knowledge base.

A full-mark answer, built from the mark scheme (not a real candidate's):

The knowledge base holds the facts about the subject area of the expert system.¹ A human expert in that subject supplied these facts.² When a user enters a query, the inference engine interrogates the knowledge base.³ The inference engine only uses the rules that apply to that query.⁴

Where the marks come from ¹"Knowledge base holds facts (1)". ²"gathered from an expert (1)". ³"It is interrogated by the inference engine (1st)". ⁴"Using relevant rules/rules that apply (1)". (Mark scheme 9626/11, November 2025, p. 5)

A weak answer of the kind the examiners describe (written for this guide, about 1-2 of 4):

The knowledge base stores information and rules. The expert system uses it to make decisions.

'Stores rules' matches a mark scheme point. 'Stores information' may get the 'holds facts' mark. The answer does not say that the facts come from an expert. It does not say that the inference engine interrogates the knowledge base. It also does not say that the inference engine uses only the relevant rules.

"For example, whilst many candidates stated that the knowledge base holds facts, very few stated that these were sourced from experts. Similarly, candidates were aware that rules are used to interrogate the knowledge base, but very few stated that only the relevant rules were used" (Examiner report, November 2025, Paper 9626/11, p. 2).

Worked example

Q4(a) [4 marks] "The directors of a company wish to create a database of customers. They are not sure whether they should ask a computer programmer to produce custom written software or they should purchase off-the-shelf software. (a) Describe the features of custom written software."

Question paper 9626/13, June 2022, p. 4

A full-mark answer, built from the mark scheme (not a real candidate's):

A programmer develops custom written software specially for one specific company, here the directors' company.¹ The programmer makes it to fit that company's particular preferences and needs, such as the exact customer details the directors want to store.² The company gets support directly from the programmers who created it.³ The company that commissions the software has to pay the development costs.⁴

Where the marks come from ¹"Software that is specially developed for a specific company/business". ²"It is made to accommodate that customer's particular preferences and needs". ³"Support is obtained directly from the creators". ⁴"Development costs have to be paid for by client for whom its written/ commissioning company/business".

(Mark scheme 9626/13, June 2022, p. 4)

A weak answer of the kind the examiners describe (written for this guide, about 0 of 4):

Custom written software is more expensive. It is better than off-the-shelf software.

'More expensive' is a vague comparison with no detail. 'Better' is an opinion and names no feature. The cost point earns a mark only when it says who pays the development costs.

"Some weaker responses contained responses such as 'more expensive' which is not enough at this level and statements such as this should always be expanded" (Examiner report, June 2022, Paper 9626/13, p. 8).

Explain

Explain. Cambridge's definition: "set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence" Syllabus 2025 to 2027, p. 66

Statement, then reason

1. An explanation has two parts. First, you make a statement. Then you give a reason, a purpose or more detail about that statement.

"An explanation requires a statement or description followed by a reason or purpose or further expansion of that statement" (Examiner report, June 2024, Paper 9626/32, p. 23). *"An explanation is statement of fact with a reason or expansion giving its purpose"* (Examiner report, March 2024, Paper 9626/32, p. 8).

2. Join the statement to its reason with a linking word such as 'so', 'so that', 'because' or 'which means'. The mark schemes and examiner reports put these words where the reason starts. Write the linking word in your answer too.

"The lower the sample rate, the smaller the file size (1st) – because there is less data stored (1)" (Mark scheme 9626/12, March 2024, p. 9). *"stating the task 'to check that the project has ended' should have been followed by a reason such as 'so that all stakeholders and clients can agree that no further additions or changes are necessary'"* (Examiner report, June 2025, Paper 9626/33, p. 25).

3. Write explanations in full sentences. A sentence keeps the point and its reason together.

"Candidates should be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences" (Examiner report, November 2022, Paper 9626/31, p. 12).

Where the marks go

1. Some Explain mark schemes label a statement (1st) and the reason or detail under it (1). The (1) mark is only for the reason or detail you add to that statement. The (1st) label works as in the general rules at the front. 176 Explain parts appeared from 2022 to March 2026, and 174 have a mark scheme. 38 of those 174 mark schemes use (1st).

"The sampling rate can affect the quality of audio files/how the audio sounds (1st) – because it captures sounds at discrete points (and not the whole sound) (1)" (Mark scheme 9626/12, March 2024, p. 9). *"Access control can be based on attribute of user within company (1st) – so that they can only access data appropriate for their job/task/role (1)"* (Mark scheme 9626/32, March 2024, p. 7).

2. A statement with no reason can still get its one mark, and no more. If every point you write is only a statement, the mark scheme can stop you getting full marks.

"A less detailed answer, such as 'they can learn at their own pace' can be given credit as it is a correct statement of the effect, but it lacks the detail to gain any further credit because it does not add an expansion" (Examiner report, June 2024, Paper 9626/32, p. 23). *"MAX two marks if answers are all only first marks"*

3. Sometimes the mark scheme links a reason or detail to a first point. The reason then earns its mark only if you also make that first point. In one encryption question, the two further marks went only to answers that first said encryption scrambles the data. In a 2-mark question, one full point and its reason can be the whole answer.

"A relatively easy first mark was given for any answer that stated that the data is scrambled. Many candidates achieved this mark. Two further marks were then dependant on this answer" (Examiner report, June 2024, Paper 9626/11, p. 4).

"Max two for one complete explanation" (Mark scheme 9626/12, March 2024, p. 9).

Reasons that fit the question

1. Explain asks why or how. Examiner reports say that many answers described what something is and lost marks.

"Most responses gave descriptive answers rather than explanations"

(Examiner report, November 2023, Paper 9626/12, p. 6). *"there is usually little or no credit given for describing what the point or process is since this does not answer the question 'how'"*

(Examiner report, November 2022, Paper 9626/31, p. 12).

2. A common mistake is to leave out the reason. A short phrase such as 'makes the code more readable' does not get full marks.

"A common mistake was to omit a reason or expansion to the response – the question asked for explanations"

(Examiner report, November 2023, Paper 9626/32, p. 16). *"a significant number only stated that comment make the code more readable so did not gain the full marks"* (Examiner report, March 2024, Paper 9626/32, p. 8).

3. Use exact technical words. In one encryption question, many answers said encrypted data 'cannot be understood'. A similar number said it 'cannot be read'. Only 'cannot be understood' got the mark.

"Many candidates made the point that encrypted data can not be understood, but an approximately equal number stated that encrypted data could not be read, and were not awarded marks"

(Examiner report, June 2024, Paper 9626/11, p. 4).

4. If the question has a scenario, use it. Give reasons that fit the situation in the question. General facts alone are not enough.

"As with all scenario-based questions, candidates are advised to consider the significance of this scenario"

(Examiner report, November 2023, Paper 9626/13, p. 9). *"The full range of marks is only available to candidates for answers referring to the scenario in the questions"* (Examiner report, November 2022, Paper 9626/31, p. 12).

Worked example

Q3(c) [3 marks] "Mainframe computers were developed from the need for computers to process data in tasks such as carrying out a census. (c) Explain why mainframe computers are used to process the data from a census." Question paper 9626/13, November 2023, p. 3

A full-mark answer, built from the mark scheme (not a real candidate's):

A census collects a very large amount of data that the computer has to process,¹ as countries tend to have a lot of people and each person adds more data.² The computer also needs to process all of this data in a short time, so a mainframe with a high processing speed does the work.³

Where the marks come from ¹ "The amount of data that has to be processed is very large (1st)". ² "as countries tend to have a lot of people (1)". ³ "All data needs to be processed in a short period of time (1st)".

(Mark scheme 9626/13, November 2023, p. 4)

A weak answer of the kind the examiners describe (written for this guide, about 2 of 3):

There is a lot of data in a census. Mainframes are very reliable.

Both sentences are only first marks, with no reason after them. The mark scheme gives a maximum of 2 marks when all the answers are only first marks. The examiners also said that most answers stopped at 'a large amount of data'.

"Typically, however, responses tended to focus on that fact that there is usually a large amount of data"
(Examiner report, November 2023, Paper 9626/13, p. 9).

Worked example

Q10(a) [3 marks] "Explain how each of these methods keeps data safe from unauthorised users. (a) encryption" Question paper 9626/11, June 2024, p. 11

A full-mark answer, built from the mark scheme (not a real candidate's):

Encryption scrambles the data,¹ so that anyone who intercepts it cannot understand it,² which means the data is of no use to a hacker or any other third party.³

Where the marks come from ¹"Data is scrambled (1st)". ²"so it is not understandable (1)". ³"of no use to a hacker/third party (1)". (Mark scheme 9626/11, June 2024, p. 9)

A weak answer of the kind the examiners describe (written for this guide, about 0 of 3):

Encryption makes the data unreadable so hackers cannot read it.

It does not say that encryption scrambles the data. So the first mark is missing, and the two marks that depend on it are lost too. 'Cannot be read' is the wrong idea. The examiners wanted 'cannot be understood'.

"Rather than making data unreadable, encryption puts data into a state whereby it can not be understood"
(Examiner report, June 2024, Paper 9626/11, p. 4).

Discuss

Discuss. Cambridge's definition: "write about issue(s) or topic(s) in depth in a structured way"

Syllabus 2025 to 2027, p. 66

How the marks are given

1. Marks go to points from a list in the mark scheme. A point is often worth 1 mark. You get a second mark when you expand it. The mark scheme shows the point as (1st) and the expansion as (1). A point with no expansion loses marks.

"Direct data source/direct data is reliable/trusted (1st) – because we collected it (1)"

(Mark scheme 9626/12, June 2024, p. 7). *"Orders are/can be processed straight away (1st) • so process of filling/completing orders can start sooner (than for batch processing) (1)"* (Mark scheme 9626/11, November 2023, p. 9).

"Must have expansions or comparisons to be a proper discussion" (Mark scheme 9626/12, March 2022, p. 4).

2. Each side often has a maximum. We checked 77 Discuss parts in the 2022-2025 mark schemes. About 36 of them print a separate MAX for each side. For 8-mark questions it is usually MAX 6 (sometimes 5, once 7). On those questions, an answer with only benefits cannot get full marks.

"This method gathers direct data, which is more up to date (1) Benefits max 6:"

(Mark scheme 9626/12, November 2024, p. 9). *"Disadvantages MAX five from"* (Mark scheme 9626/12, June 2023, p. 7).

"Content areas could include: Max SIX from: Benefits: e.g.:" (Mark scheme 9626/31, June 2025, p. 12).

3. The mark scheme caps one-sided answers. We checked 77 mark schemes. 23 of them (all 2022-2023) say you need at least two points on each side for full marks. On some 4-mark and 6-mark questions, one of each is enough. 4 more mark schemes (November 2023) cap an answer with only benefits or only drawbacks. The 2023 and 2025 examiner reports say the same thing. You cannot get full marks without both sides.

"Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks"

(Mark scheme 9626/12, March 2022, p. 4). *"Answers that do not provide both do not access the highest marks because a proper discussion should look at both sides of an issue"* (Examiner report, June 2023, Paper 9626/32, p. 22). *"Also, where both benefits and drawbacks are required by the question, answers that address only benefits or only drawbacks cannot achieve full marks"* (Examiner report, June 2025, Paper 9626/32, p. 22).

4. Sometimes the mark scheme gives a mark for a short opening definition or description of the thing in the question. We found this in 18 of the 77 mark schemes. Usually the cap is one mark for each thing you describe. Once, in June 2025 Paper 31 Q7, it was two marks. So one clear opening sentence can help. A long introduction cannot earn more.

"One mark from: • Definition of encryption (1) Max 6 from: Benefits:" (Mark scheme 9626/12, June 2025, p. 11). *"Max one for: • Definition of computer simulation (1) Max six from: Benefits"* (Mark scheme 9626/11, November 2025, p. 9). *"Eight from: Simple statement of the difference between the two (1): e.g"* (Mark scheme 9626/12, June 2024, p. 7).

5. No 2022-2025 Discuss mark scheme we checked gives a mark for a conclusion or for your opinion. The current syllabus meaning of Discuss does not ask for one. Older examiner reports (2019) used an older wording. It said Discuss often needs a conclusion. The syllabus no longer defines Discuss that way. A one-line conclusion does no harm. But spend your time on expanded points.

"Discuss write about issue(s) or topic(s) in depth in a structured way" (Syllabus 2025 to 2027, p. 66). *"Discuss questions require candidates to give the important arguments for and against, and often require a conclusion"*

(Examiner report, November 2019, Paper 9626/12, p. 4).

How to write it

1. Bullet points lose marks. Every 2022 and 2023 Discuss mark scheme caps a bulleted list. So do 6 of the 17 mark schemes in 2024. The caps are max 6 of 8, max 4 or 5 of 6, and max 3 of 4. In total, we counted 48 of the 77 mark schemes with this cap.

"Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion"

(Mark scheme 9626/12, March 2022, p. 4). *"Must be at least one of each for full marks. Max 4 marks if bullets/list of points"* (Mark scheme 9626/32, March 2023, p. 11).

"Must have at least one of each to gain full marks Max. three marks if bullets/list of points" (Mark scheme 9626/12, November 2022, p. 8).

2. The 2025 mark schemes no longer print the bullet cap. But the examiners still say that lists and tables of statements cannot reach most of the marks. This also applies to two separate lists (benefits, then drawbacks) made of single words or very short sentences.

"so learners who produced bulleted lists or tables of statements were unable to access most of the marks"

(Examiner report, March 2025, Paper 9626/32, p. 10). *"The command word 'discuss' requires candidates to provide a structured answer, so a bulleted list of brief statements is not properly answering the question and does not gain access to the full range of marks"* (Examiner report, June 2024, Paper 9626/32, p. 23). *"This comment also applies to answers that have separate lists of benefits and drawbacks, but the lists consist of single word or very brief sentences"*

3. Write in full sentences and paragraphs, not lists. A good shape is one paragraph of benefits and one paragraph of drawbacks. Put the reason or effect of each point in the same sentence as the point.

"Candidates should be encouraged to write answers in full sentences"

(Examiner report, June 2025, Paper 9626/31, p. 17). *"Questions that ask candidates to discuss the benefits and drawbacks of something require responses to give both sides of the argument and should be in continuous prose rather than bullet points"* (Examiner report, March 2022, Paper 9626/12, p. 1). *"The strongest answers were written in full prose style and included points which were expanded upon"* (Examiner report, March 2022, Paper 9626/12, p. 2).

What goes wrong

1. Making a point and stopping. Examiners say most students give correct points, but few students expand them. Two 2022 mark schemes cap answers with no expansion at 6 of 8.

"Most responses gained credit for making valid points, but few went on to discuss the points they made by expanding on them" (Examiner report, March 2022, Paper 9626/12, p. 2). *"Max. six marks if bullets/list of points Max six marks if no attempt at expansion"* (Mark scheme 9626/12, November 2022, p. 10).

2. Using vague words about cost. The examiners said that 'expensive' or 'costs more' was too vague. Say what the money pays for, for example travelling to interviews or printing questionnaires.

"Where candidates stated that collecting direct data was expensive, this was considered too vague, but if candidates identified an area on which money would have to be spent, which would not otherwise be spent, this was awarded" (Examiner report, June 2024, Paper 9626/12, p. 7). *"Many responses mentioned costs without going into sufficient detail to merit credit"* (Examiner report, March 2022, Paper 9626/12, p. 2).

3. Writing the same point twice: once as a benefit, then reversed as a drawback. The examiner gives the mark only once, so you waste time.

"Some candidates effectively wrote what was good about direct data compared to indirect data and then repeated the list, but in reverse structure, so the positive aspects of one became the negative points of the other" (Examiner report, June 2024, Paper 9626/12, p. 7). *"(Converse answer may also be given as an advantage – Award ONCE only)"* (Mark scheme 9626/12, June 2023, p. 7).

4. Describing the topic instead of discussing it in the setting of the question. Some answers described types of technology or the stages of a process. Others wrote about the wrong topic. These answers got few or no marks. Link each benefit and drawback to the people or organisation named in the question.

"Descriptions of different types of TEL, statements about benefits or drawbacks gained little or no credit without further comment on how these can affect the achievement of students" (Examiner report, March 2025, Paper 9626/32, p. 10). *"there were many who made mistakes such as describing the stages of the data mining process or confusing data mining with another topic altogether"* (Examiner report, June 2025, Paper 9626/31, p. 19). *"Answers should have focused on how banking organisations make use of data mining"* (Examiner report, June 2025, Paper 9626/32, p. 23).

Worked example

Q7 [8 marks] "Data can be collected by using a direct or an indirect data source. Discuss the advantages and disadvantages of using direct data compared to indirect data."

Question paper 9626/12, June 2024, p. 8

A full-mark answer, built from the mark scheme (not a real candidate's):

The person who will use direct data collects it first-hand (a primary source). Someone else collected indirect data for another purpose (a secondary source).¹ One advantage of direct data is that it is more reliable and trusted, because we collected it ourselves and know exactly how we gathered it.² Also, we collect only specific, relevant data, so we do not waste time sorting through data we do not need.³

Direct data is also more up to date than indirect data, which someone may have collected years ago.⁴ However, a disadvantage is that the sample size may be small, whereas indirect data sources tend to be much larger sets of data.⁵ Collecting direct data also costs money that we would not otherwise spend, such as travelling to interview people and printing questionnaires.⁶

Where the marks come from ¹ "Simple statement of the difference between the two (1): e.g". ² "Direct data source/direct data is reliable/trusted (1st) – because we collected it (1)". ³ "ONLY Specific/relevant data is collected (1)". ⁴ "Direct data is (more) up-to-date (1) Disadvantages MAX SIX". ⁵ "The sample size may be small (1st) – whereas indirect data sources tend to be larger sets of data (1)". ⁶ "Collecting direct data involves the expense of travelling/photocopying/printing questionnaires//any expense to do with a direct data method (1)".
(Mark scheme 9626/12, June 2024, p. 7)

A weak answer of the kind the examiners describe (written for this guide, about 2 of 8):

Advantages: direct data is cheaper, relevant, up to date. Disadvantages: indirect data is not relevant and old, direct data is expensive.

It is a list, not a discussion. 'Expensive' does not say what the money pays for. The disadvantages only repeat the advantages in reverse, so they earn nothing new.

"Some candidates effectively wrote what was good about direct data compared to indirect data and then repeated the list, but in reverse structure, so the positive aspects of one became the negative points of the other"
(Examiner report, June 2024, Paper 9626/12, p. 7).

Six sentences earn eight marks because two of them give a point plus its expansion. The examiner report says this question was about the good and bad sides of direct data. Indirect data was only the comparison. So do not write each point twice in reverse.

Worked example

Q7 [8 marks] "Discuss the benefits and drawbacks of using data mining in businesses."

Question paper 9626/31, June 2025, p. 8

A full-mark answer, built from the mark scheme (not a real candidate's):

Data mining is the use of software tools to analyse very large data sets.¹ Its aim is to discover trends, patterns and relationships in the data that people would not spot by hand.² Data mining can improve sales and profits, because the business can target its advertising at the types of customer most likely to buy.³ The information it extracts also gives managers a sound basis for their decisions, instead of guesswork.⁴ On the other hand, the extracted information may not be accurate, because it depends on how good the source data set is.⁵ The business must also protect the privacy of the individuals in the data and follow data protection laws, which adds work and risk.⁶

Where the marks come from ¹ "data mining (is a set of techniques and tools) used to from work on / analyse large data sets". ² "to discover trends / patterns / to find relationships". ³ "used to improve sales / profits (1st) by targeting customers / demographics (1)". ⁴ "provides a sound basis for decision-making". ⁵ "extracted information may not be accurate (1st) depends on source of data set (1)". ⁶ "must ensure privacy of individuals / conform to data protection regulations". (Mark scheme 9626/31, June 2025, p. 12)

A weak answer of the kind the examiners describe (written for this guide, about low (2-3 of 8):

A two-column table: Benefits | Drawbacks; trends | expensive; profit | privacy; decisions | storage.

Single words in a table name points, but they give no reasons or effects. So there is no discussion and no expansion marks.

"Furthermore, if these answers are shown in a sketched table, they do not properly answer the question as there is no discussion" (Examiner report, June 2025, Paper 9626/31, p. 17).

This 2025 mark scheme gave up to two marks for saying what data mining is. Then it gave MAX SIX for each side. A two-sentence definition, two benefits and two drawbacks reach eight marks, if you expand one of each with its reason.

Evaluate (and Assess)

Evaluate. Cambridge's definition: "judge or calculate the quality, importance, amount, or value of something"

Syllabus 2025 to 2027, p. 66

Assess. Cambridge's definition: "make an informed judgement" Syllabus 2025 to 2027, p. 66

How the marks are given

1. You must write about both sides. Some 2022-2023 mark schemes cap answers that are all for or all against at 6 of 8. This is 3 of the 24 Evaluate mark schemes from 2022 on. Two mark schemes say you need at least two points on each side for full marks.

"Max 6 if all for or all against. Max 6 marks if bullets/list of points" (Mark scheme 9626/32, November 2023, p. 10). *"Must be at least two of each for full marks. Max 6 marks if bullets/list of points"* (Mark scheme 9626/32, March 2023, p. 7). *"Must be at least 2 strengths and at least 2 weaknesses for full marks"* (Mark scheme 9626/32, November 2022, p. 6).

2. Recent mark schemes give one mark for a conclusion. We checked the Evaluate parts from 2022 onward. 8 of the 24 with a mark scheme have a conclusion mark. All 8 are from November 2024 or later. That is 8 of the 9 mark schemes from that date on. End with a short overall judgement that brings your points together.

"Valid conclusion drawing together/based on arguments presented in the answer (1)" (Mark scheme 9626/13, June 2025, p. 9). *"Max one from: Conclusion: • One mark for a reasoned conclusion / overall judgement"* (Mark scheme 9626/32, November 2025, p. 10). *"One mark is available for a valid conclusion / judgement (must have considered both pros and cons)"* (Mark scheme 9626/32, November 2024, p. 15). *"Such discussions should end with a conclusion that revisits and brings together the points made"* (Examiner report, November 2024, Paper 9626/12, p. 4).

3. Good answers take each point through three steps. First give a fact, then a consequence of that fact, then a judgement about how much it matters. Some mark schemes give a first mark for the point and more marks for the expansion. They also limit how many marks plain points can earn.

"Good answers gave a description of how compression was used or of the effects of the compression and then added a consequence followed by a judgment as to the desirability or further effect of the compression" (Examiner report, June 2025, Paper 9626/33, p. 26). *"Eight marks from: MAX four for 1st marks only Evaluations:"* (Mark scheme 9626/13, June 2025, p. 9). *"Must have expansions to be a proper evaluation"*

4. Some mark schemes give one mark for a short definition or description of the thing you evaluate. Keep it to one sentence, then start judging it.

"One mark for a description of encryption (e.g. data scrambled/converted to cyphertext)"

(Mark scheme 9626/12, March 2024, p. 9). *"Valid definition / description of video-conferencing in education"*

(Mark scheme 9626/31, June 2025, p. 15).

How to write it

1. Write in full sentences and paragraphs, not bullet points or a table. 9 of the 24 Evaluate mark schemes from 2022 on (all 2022-2024) cap a bulleted list. The cap is usually 6 of 8 marks (5 of 6 on 6-mark questions).

"A detailed answer is needed, which cannot be achieved by writing bullet points"

(Examiner report, March 2023, Paper 9626/12, p. 1). *"Must be a proper evaluation for full marks. Max 6 marks if*

bullets/list of points" (Mark scheme 9626/32, March 2022, p. 6). *"Max. five marks if all three methods not covered Max.*

five marks if bullets/list of points" (Mark scheme 9626/12, March 2023, p. 8). *"Answers with bulleted, short statements for*

free response questions, especially those requiring analyses, discussions and evaluations do not adequately answer

the question" (Examiner report, June 2025, Paper 9626/31, p. 17).

What goes wrong

1. If you leave out a conclusion, you can lose marks. In the March 2024 examiner report (Paper 12, Q9), examiners said students did not reach a realistic conclusion. This reduced their marks.

"candidates largely failed to come to any realistic conclusion about encryption, which reduced the marks scored on the question" (Examiner report, March 2024, Paper 9626/12, p. 3).

2. The most common mistake is describing how the thing works instead of judging it. Knowing the topic is not enough. You have to say whether each feature helps or causes problems.

"Candidates seemed to have some understanding of each process, but preferred to describe each of these than

actually evaluate their efficacy" (Examiner report, March 2023, Paper 9626/12, p. 3). *"In some cases, candidates*

described how control technologies worked. Typically, this involved an in-depth discussion of sensors"

(Examiner report, November 2024, Paper 9626/12, p. 3). *"candidates were expected to make factual statements about*

project management software (PMS) and then expand on these with comments about whether these facts helped or

hindered the project" (Examiner report, November 2023, Paper 9626/33, p. 21).

3. One-sided answers lose marks. Both sides must be there, but they do not need to be exactly equal. The examiners accepted more points for than against.

"some candidates seemed not to understand the importance of 'evaluate' as a command word, and so gave a one-sided

answer" (Examiner report, June 2024, Paper 9626/11, p. 1). *"Candidates tended to concentrate on the positive*

implications, rather than both the positive and the negative, and this did restrict the overall marks scored for this

question in a few cases" (Examiner report, June 2025, Paper 9626/12, p. 8). *"Typically, candidates gave more answers in*

support than against, but this was acceptable" (Examiner report, June 2025, Paper 9626/13, p. 12).

Tips

1. Judge the thing in the scenario in the question, not in general. Some answers about spreadsheets never mentioned financial modelling, and they did badly.

"To gain full credit, candidates must produce a reasoned evaluation that weighs up the advantages and disadvantages

for the given scenario/context" (Examiner report, March 2023, Paper 9626/12, p. 1). *"Whilst some lenience was allowed,*

candidates who failed to make any mention of financial modelling struggled to score well on this question"

(Examiner report, June 2025, Paper 9626/13, p. 12).

2. Evaluate only what the question names. You do not need to compare it with other methods unless the

question asks you to.

"It was not necessary to compare pilot with other possible methods of implementation because the question asked candidates to 'evaluate pilot implementation' and made no mention of the other methods"

(Examiner report, November 2023, Paper 9626/32, p. 17).

3. Evaluate is not the same as Discuss. Discuss asks you to write in depth in a structured way. Evaluate asks you to judge importance or value. You can give a judgement on each point or on the whole answer. We checked the 77 Discuss mark schemes from 2022 onward. None has a conclusion mark, but recent Evaluate mark schemes do.

"Discuss write about issue(s) or topic(s) in depth in a structured way Evaluate judge or calculate the quality, importance, amount, or value of something" (Syllabus 2025 to 2027, p. 66). "The command word 'evaluate' requires a judgement of the importance of the points being made about the method of prototyping"

(Examiner report, November 2024, Paper 9626/32, p. 17). *"In questions where an evaluation is, or evaluations are,*

required, the judgement can be for individual points or for an overall judgement on the whole content of the answer"

(Examiner report, November 2024, Paper 9626/33, p. 21).

Worked example

Q10 [6 marks] "A person uses a dialogue interface to control multiple devices in their home. Evaluate the use of a dialogue interface in this context." Question paper 9626/12, June 2025, p. 9

You control a dialogue interface by speaking to it.

A full-mark answer, built from the mark scheme (not a real candidate's):

Because the person controls the devices by speaking, they do not need to physically touch any switch or screen.¹ This is a real benefit for a disabled person, who can switch on lights or heating around the home without moving to them.² It also means the person can carry on with other tasks, such as cooking, while giving a command.³ However, the interface relies on sound, so loud music or a television in the room may stop it understanding the command.⁴ It may also mishear what the person says and operate the wrong device, so the person can rely on it less.⁵ Overall, a dialogue interface is very useful in a home, especially for someone with limited movement, if the rooms are quiet enough.⁶

Where the marks come from ¹ "No need to physically interact (may exemplify) (1st)". ² "More accessible to disabled users (may refer to device itself or devices around the home) (1)". ³ "Can undertake other tasks at the same time (1)". ⁴ "External noise may result in the dialogue interface not understanding the command (1)". ⁵ "Dialogue interface may misinterpret what is said (1)". ⁶ "One mark for a valid conclusion (1)". (Mark scheme 9626/12, June 2025, p. 9)

A weak answer of the kind the examiners describe (written for this guide, about 2 of 6):

Dialogue interfaces are good for disabled people. You control things with your voice. It is hands-free and easy to use.

Every point is positive. No point has a consequence, and there is no conclusion. So the answer cannot reach most of the marks for drawbacks and for the judgement.

"Candidates tended to concentrate on the positive implications, rather than both the positive and the negative, and this did restrict the overall marks scored for this question in a few cases"

(Examiner report, June 2025, Paper 9626/12, p. 8).

Three sentences for, two against and one conclusion. The conclusion mark is easy to get if you remember to write one.

Worked example

Q10 [8 marks] "A company is developing a data input screen for entering customer details into a database. Evaluate the use of evolutionary prototyping for developing the data input screen."

Question paper 9626/32, November 2024, p. 11

A full-mark answer, built from the mark scheme (not a real candidate's):

In evolutionary prototyping, the developers keep refining and rebuilding the data input screen until it is complete.¹ The staff who will use the screen work with every version of it during development.² As a result, the screen is more likely to have the customer fields the staff need, which matters most for a data input screen.³ The developers can also change the screen at any iteration, for example if the company decides to record extra customer details.⁴ On the other hand, the constant meetings between developers and users can be costly.⁵ This raises the overall cost of development, which is a serious drawback when the company is building only one screen.⁶ Developers may also find it hard to know when to stop tweaking the screen, so development may go on too long.⁷ Overall, evolutionary prototyping suits this screen because user feedback matters most for data entry, if the company sets a clear end point to control cost.⁸

Where the marks come from ¹"The system / screen is continually refined and rebuilt (when developed using evolutionary prototyping)". ²"The user continuously engages with the system (1)". ³"so the new system / screen is more likely to meet the user requirements (1)". ⁴"Screen / product can be changed at any stage / iteration (1)". ⁵"Continual need to consult / interact between developers / client / end-users can be costly (1)". ⁶"which raises overall cost of development (1)". ⁷"Developers may have difficulty knowing when it is necessary to stop 'tweaking' / developing the system (1)". ⁸"One mark is available for a valid conclusion / judgement (must have considered both pros and cons)". (Mark scheme 9626/32, November 2024, p. 15)

A weak answer of the kind the examiners describe (written for this guide, about at most 3 of 8):

Evolutionary prototyping involves the users. The screen can be changed. It can be expensive. It takes a long time.

These are bare statements. No statement has a consequence or a judgement after it, and there is no conclusion.

"Weaker answers confused the methods of prototyping, made statements with no consequences, expansion, or judgments" (Examiner report, November 2024, Paper 9626/32, p. 17).

Both answers contain the same facts. The strong answer earns more because it follows each fact with what it leads to and how much it matters. It also ends with a judgement.

Analyse

Analyse. Cambridge's definition: "examine in detail to show meaning, identify elements and the relationship between them" Syllabus 2025 to 2027, p. 66

How the marks are given

1. Marks come from making a point and then expanding it into what it causes or means. Some mark schemes say an answer without expansions is not a proper analysis. Two 2022 mark schemes cap it at 6 of 8. We checked the 18 Analyse mark schemes from 2022 onward. None gives a mark for a conclusion.

"Max six marks if bullets/list of points Max six marks if no attempt at expansion"

(Mark scheme 9626/13, November 2022, p. 6). *"Must be a proper analysis to gain full marks Max. six marks if bullets/list of points Must have expansions to be a proper analysis"* (Mark scheme 9626/11, June 2022, p. 4). *"candidates should produce answers that make a point and then expand on the point"* (Examiner report, June 2023, Paper 9626/32, p. 22).

2. Show the links between the parts: one thing leads to another, which leads to a further effect. Examiners and mark schemes reward these chains of cause and effect.

"Good answers should have made references to the need for greater bandwidth, for example, so IT infrastructure has to be upgraded which increases costs taking money from other needs" (Examiner report, June 2023, Paper 9626/31, p. 18).

"Increased digital awareness leads to increased/better infrastructure// more investment in IT (accept personal equivalences) (1) which then leads to increased digital awareness (1)" (Mark scheme 9626/12, March 2024, p. 10).

"because at each compression/decompression more image information is lost (1)"

(Mark scheme 9626/32, November 2023, p. 5).

3. A short description or definition at the start can earn one or two marks, but no more. A description only earns marks when you use it to explain the effect or the reason.

"Max two from: Resolution is measured/described as pixels per inch/ppi" (Mark scheme 9626/32, March 2022, p. 5).

"Credit was given for a brief description of UHD television but the command word (see. p. 45 of the syllabus) analyse requires candidates to 'examine in detail to show meaning, identify elements and the relationship between them' so this question demanded more" (Examiner report, June 2023, Paper 9626/31, p. 18). *"Descriptions of vector images were not given credit unless these descriptions were used to explain why vector graphics are, or are not, used on websites"*

(Examiner report, March 2025, Paper 9626/32, p. 12).

How to write it

1. Write in full sentences. 10 of the 18 Analyse mark schemes from 2022 on (all 2022-2023) cap bullet points. The caps are 6 of 8, 5 of 7, or 4 or 5 of 6.

"questions that ask candidates to 'analyse' or 'discuss' require detailed responses that include a proper analysis or discussion, rather than short or bullet-pointed responses" (Examiner report, March 2023, Paper 9626/12, p. 1). *"Max 4 marks if bullets/list of points/characteristics"* (Mark scheme 9626/31, June 2023, p. 8). *"Max. six marks if bullets/list of points Max six if no expansions"* (Mark scheme 9626/11, November 2022, p. 6).

What goes wrong

1. Writing about what the thing is or how it works, instead of the effect the question asks about. If the question asks about risks to an economy or effects on an image, every point must reach that.

"This question was not about digital currencies and what they are or how they work. It was about the risks to an economy that might occur as a result of the use of digital currencies" (Examiner report, June 2023, Paper 9626/32, p. 22).

"references to a reduction in file size without an accompanying consequence on the image itself did not answer the question" (Examiner report, November 2023, Paper 9626/32, p. 16).

2. Listing features. A list of points without expansion gets few marks.

"Responses often consisted of lists of the features of encryption" (Examiner report, June 2022, Paper 9626/11, p. 2). *"The weaker answers did not go beyond listing points so could not gain many marks"*

3. Ignoring the focus of the question, or being vague. Stay on the exact topic in the question. Avoid empty phrases like 'looks better'.

"Many candidates attempted to give answers that did not focus on the production of weather forecasts, or were too vague and little credit could be given" (Examiner report, November 2023, Paper 9626/13, p. 10). *"Questions about the digital divide require candidates to concentrate on the specific focus, otherwise answers tend to be generic, unfocused and, usually, largely irrelevant"* (Examiner report, March 2024, Paper 9626/12, p. 3). *"Vague references to 'vector graphics make the site look better' or 'look attractive to students' did not gain credit"*

(Examiner report, March 2025, Paper 9626/32, p. 12).

Tips

1. Cover good and bad sides where they exist. You do not always need both for full marks. But some mark schemes cap answers that cover only one side or one item.

"While it was possible to gain full credit without doing so, a full analysis should have given details of both 'for' and 'against' the use of the waterfall method in the question scenario" (Examiner report, June 2024, Paper 9626/32, p. 23).

"Typically, some of the stronger responses did detail the good and the bad aspects and explain them in detail to show meaning" (Examiner report, June 2022, Paper 9626/32, p. 18). *"Max 4 for all lossless or for all lossy"*

(Mark scheme 9626/32, November 2023, p. 5).

2. Read the whole question. If it names something and asks about 'other' ones, the named ones earn nothing.

"a significant number of a candidates included the tools mentioned in the question in their answer, which did not gain them credit" (Examiner report, June 2025, Paper 9626/32, p. 20).

Worked example

Q3 [6 marks] "Bitmap images can be compressed when stored and used on computer systems. Analyse the effects of the two main methods of image compression on a bitmap image."

Question paper 9626/32, November 2023, p. 4

A full-mark answer, built from the mark scheme (not a real candidate's):

The two main methods of image compression are lossless compression and lossy compression.¹ Lossless compression stores all the image information exactly, because it removes no image information.² As a result, the computer can recreate the image perfectly, with no difference from the original.³ Lossy compression loses information during compression, and you cannot get that information back.⁴ Because of this, the image loses quality and may look blurred or blocky.⁵ The quality gets worse each time you compress and save the image again, so an image that someone edits many times keeps losing detail.⁶

Where the marks come from ¹"(The two methods are) lossless compression and lossy compression". ²"All image information is stored exactly/no image information is removed". ³"Image is/can be recreated perfectly/with no difference from the original". ⁴"Information is lost during compression and is not/cannot be replaced". ⁵"Images are degraded/reduced in quality/blurred/pixelated/blocky". ⁶"Image quality is further degraded/reduced with repeated compression/decompression/saving and opening (1st)". (Mark scheme 9626/32, November 2023, p. 5)

A weak answer of the kind the examiners describe (written for this guide, about at most 1 of 6):

Compression makes the file size smaller, so the image takes up less storage space and is quicker to send by email. Lossy makes the file smaller than lossless.

It talks about file size and sending time, not about the effect on the image. The question asked about the effect on the image.

"references to a reduction in file size without an accompanying consequence on the image itself did not answer the question" (Examiner report, November 2023, Paper 9626/32, p. 16).

The answer covers both methods (the mark scheme allows at most 4 marks for one method). Each sentence says what happens to the image. The 'because' and 'as a result' links are the analysis.

Justify

Justify. Cambridge's definition: "support a case with evidence/argument" Syllabus 2025 to 2027, p. 66

How the marks are given

1. Most marks go to the reasons. About half the Justify mark schemes (9 of 19 from 2022 on) give a first mark for a benefit. They give a second mark for the reason or result that follows from it.

"It is the reasons that gain most credit in questions that ask for 'justify'"

(Examiner report, November 2024, Paper 9626/33, p. 19). *"Give a positive impact (1st) • Give an extension that is a positive implication on a customer or business of the identified justification. (1)"*

(Mark scheme 9626/13, November 2024, p. 8). *"One mark per valid validation check identified (max 5) and justification given to a maximum of eight marks"* (Mark scheme 9626/12, March 2025, p. 7).

2. Some mark schemes give one mark for a definition of the thing you are justifying. One clear sentence is enough.

"Any correct definition of direct data sources, such as:" (Mark scheme 9626/11, June 2023, p. 10). *"Definition of*

encryption (1) • Protects data/data "is safe"/improves data security (1)" (Mark scheme 9626/12, November 2025, p. 8).

"Definition of online processing: ONE from: • User is in direct connection with a central computer (1)"

(Mark scheme 9626/13, November 2024, p. 8).

How to write it

1. Write in full sentences. Four 8-mark Justify mark schemes (all 2023) cap a bulleted list at 6 marks.

"so less expensive media/data storage costs (1) Max 6 marks for just bullet points"

(Mark scheme 9626/13, November 2023, p. 9). *"indirect data is not hers to sell/pass/share (1) MAX 6 marks for a bulleted*

list" (Mark scheme 9626/11, June 2023, p. 10). *"Candidates should be discouraged from writing bulleted, short statements in their responses but instead be encouraged to write answers in full sentences"*

(Examiner report, March 2024, Paper 9626/32, p. 7).

What goes wrong

1. Justify is one-sided. Arguments against the choice earn nothing. One examiner report (June 2025) says no question that uses Justify needs drawbacks. Several examiner reports describe students who gave 'for and against' answers. These answers did not meet the demand of the question.

"Some responses did not correctly interpret the command word of 'justify', and so gave arguments for and against. To justify an action, responses need to support an argument rather than criticise"

(Examiner report, November 2023, Paper 9626/12, p. 7). *"Most responses also gave arguments against the decision. These answers did not meet the demand of the question and were not worthy of credit"*

(Examiner report, November 2023, Paper 9626/13, p. 10). *"Drawbacks of using mail merge were not required when answering this question, nor are they required for any question that uses the command word 'justify'"*

(Examiner report, June 2025, Paper 9626/33, p. 25).

2. Listing features or describing how something works is not justifying. Turn each feature into a reason for using it.

"Most candidates could list the features of a VPN, but few used these to justify its use"

(Examiner report, June 2025, Paper 9626/33, p. 26). *"many candidates ignored the command word and wrote, it appeared, all they knew about lossy, and lossless, compression rather than stating facts and using these as reasons why lossy compression would be used"* (Examiner report, March 2024, Paper 9626/32, p. 7). *"few candidates attempted to fully justify, with some candidates giving reasons against the decision, and others giving descriptions"*

(Examiner report, November 2025, Paper 9626/11, p. 3).

3. Repeating the same benefit in different words does not earn extra marks. Give a range of different reasons.

"Whilst candidates were aware of some advantages, these were often narrow and repeated from different angles in an apparent attempt to achieve more marks" (Examiner report, November 2024, Paper 9626/13, p. 9). *"To justify the use of an interpreter, candidates had to have a range of benefits of the use of an interpreter"*

(Examiner report, June 2025, Paper 9626/13, p. 11).

4. Vague reasons lose marks. Be technically precise about why the choice helps.

"some of the reasons given in support were either vague or incorrect"

(Examiner report, November 2025, Paper 9626/12, p. 7). *"In some cases, there was a lack of technical understanding, whilst in others, answers were vague and lacked clarity"* (Examiner report, June 2025, Paper 9626/13, p. 11).

Tips

1. You may mention a drawback of an alternative, because that still supports the choice. For example: interviews give data relevant to the study, but indirect sources are not specific to it.

"The command word 'justify' requires answers to support a case so drawbacks do not gain credit unless they are of alternatives" (Examiner report, June 2024, Paper 9626/31, p. 20). *"This justification may, for example, include why an alternative method has not been adopted"* (Examiner report, June 2023, Paper 9626/13, p. 11). *"You are in control of the sample size//can be as big/small as needed (1st) – whereas Isabella has no control over the amount of data collected from an indirect source (1)"* (Mark scheme 9626/11, June 2023, p. 10).

2. Link every reason to the scenario. Full marks may only be possible when you combine the concept and the context. A check or feature that does not fit the given data earns nothing.

"The mark scheme was structured so that full marks could only be achieved where the concept and the context were combined" (Examiner report, November 2025, Paper 9626/12, p. 7). *"In questions such as this, context is important. For example, a student number is unlikely to be validated with a range check, as candidates were not given a range"* (Examiner report, March 2025, Paper 9626/12, p. 3). *"Candidates should apply their knowledge to the scenario in the question set because the full range of marks is only available to candidates for answers referring to the question"*

3. If the question names two things to justify, cover both. One mark scheme capped answers that dealt with only one of them at 6 of 8.

| "Max six if only one of student number or examination score used" (Mark scheme 9626/12, March 2025, p. 7).

Worked example

Q9 [4 marks] "A programmer is developing a new piece of software. The programmer has decided to write code that needs to be interpreted so that it will run. Justify the programmer's decision."

Question paper 9626/11, November 2025, p. 7

A full-mark answer, built from the mark scheme (not a real candidate's):

With an interpreter, the programmer can write the code in a high-level language, which is easier to write and read.¹ This means the programmer does not need to code in machine code.² The same program code will run on different computers and operating systems, as long as each has an interpreter.³ The interpreter reports mistakes as soon as it finds them, so the programmer can debug the new software quickly while developing it.⁴

Where the marks come from ¹"Programmer can write in high level language (1)". ²"Programmer does not need to code in machine code (1)". ³"The program code will run on different computers/operating systems (1)". ⁴"An interpreter will highlight mistakes as they are found (1)". (Mark scheme 9626/11, November 2025, p. 8)

A weak answer of the kind the examiners describe (written for this guide, about 1 of 4):

Interpreted code is easier to debug. However, it runs more slowly than compiled code and the user needs the source code, so a compiler would be better.

Only the first sentence supports the decision. The rest argues against it, which earns nothing for Justify.

| "few candidates attempted to fully justify, with some candidates giving reasons against the decision, and others giving descriptions" (Examiner report, November 2025, Paper 9626/11, p. 3).

| Four marks need four separate reasons in favour. Do not mention compilers unless you use them to show why the interpreter is better.

Worked example

Q10 [8 marks] "The manager of the payroll department within a company has commissioned a computer programmer to write software to run the payroll. Justify this decision to use custom written software." Question paper 9626/13, November 2023, p. 11

A full-mark answer, built from the mark scheme (not a real candidate's):

The programmer creates custom written software for this exact purpose: running this company's payroll.¹ So the payroll staff can use it without adapting it first.² The programmer can make sure the software has all the features the manager requires, such as the company's own pay rules.³ It will not include unnecessary features, so payroll staff are not distracted by bloatware.⁴ The programmer can

change any settings built into the software.⁵ This matters if the software does not meet the manager's requirements at first.⁶ The company will also own the copyright of the payroll software.⁷ This means it could sell the software to other businesses and earn money from it.⁸

Where the marks come from ¹"(This) custom written software is created for a defined/purpose/running a payroll". ²"Does not need to be adapted for use (1)". ³"Software will have/Programmer can ensure the software will have all the features that are required (1)". ⁴"Custom written software will not have any unnecessary/distracting features/bloat ware (1)". ⁵"(Any) settings built into the software can be changed (1st)". ⁶"if the software doesn't initially meet the manager's requirements (1)". ⁷"The company will own the copyright of the custom written software (1st)". ⁸"so can sell it to others (1)". (Mark scheme 9626/13, November 2023, p. 9)

A weak answer of the kind the examiners describe (written for this guide, about 2 of 8):

Custom software is made for the company, so it meets their needs. However, it is expensive and takes a long time to write, and it may have bugs because few people have tested it. Off-the-shelf software is cheaper and ready straight away.

Only the first sentence supports the decision. The drawbacks of custom software and the benefits of off-the-shelf software argue against the manager's choice. So they earn nothing.

"Most responses also gave arguments against the decision. These answers did not meet the demand of the question and were not worthy of credit" (Examiner report, November 2023, Paper 9626/13, p. 10).

Every sentence argues for the manager's decision. Where a mark scheme shows (1st) then (1), state the benefit and then say what it leads to.

Compare and Contrast

Compare. Cambridge's definition: "identify/comment on similarities and/or differences"

Syllabus 2025 to 2027, p. 66

Contrast. Cambridge's definition: "identify/comment on differences" Syllabus 2025 to 2027, p. 66

Which sides to give

1. Mark schemes for Compare list similarities and differences, with one mark for each valid point. Some mark schemes need at least one similarity and one difference for full marks. Two examples are March 2023 Paper 32 and June 2022 Paper 12.

"Must be 1 of each for full marks" (Mark scheme 9626/32, March 2023, p. 5). "Must have at least one similarity and one difference to gain full marks" (Mark scheme 9626/12, June 2022, p. 4).

2. Some mark schemes cap each side. A 5-mark question can allow at most 4 marks for similarities and 4 for differences. An answer with one side only cannot reach full marks.

"Five marks available: Similarities Max 4" (Mark scheme 9626/11, November 2023, p. 5). "Five from: Max 4 from: Similarities" (Mark scheme 9626/32, November 2025, p. 6).

3. Give similarities and differences in a Compare answer, even when the question does not ask for both. The examiner reports differ here. The March 2023 report advised both sides in every case. The November 2022 report gave full marks for one side on one question. An answer with both sides is safe

under either view.

"even if it does not, candidates are advised to give both to ensure access to the full mark range"

(Examiner report, March 2023, Paper 9626/32, p. 8). *"However, in this case, full marks could have been achieved with only similarities or only differences in the answers"* (Examiner report, November 2022, Paper 9626/32, p. 18).

4. Students often leave out the similarities. The November 2025 examiner report (Paper 11) said few, if any, candidates gave similarities. The June 2025 report (Paper 32) said candidates often omitted them.

"Few, if any, candidates identified similarities between the two methods"

(Examiner report, November 2025, Paper 9626/11, p. 2). *"the similarities between the two were often omitted"*

(Examiner report, June 2025, Paper 9626/32, p. 22).

5. Contrast means differences only. Similarities in a Contrast answer get no marks and use up your time.

"Any references to similarities were ignored and not given credit because these did not answer the question"

(Examiner report, June 2023, Paper 9626/32, p. 21). *"Command word: Contrast: identify/comment on differences"*

(Mark scheme 9626/32, March 2022, p. 3).

Write each point as a comparison

1. Make each point say something about both things. 'A LAN covers a small area' gets no mark. 'A LAN covers a smaller area than a WAN' gets a mark. The June 2025 report on a Contrast question said you need not write out both sides every time. The point must still read as a comparison.

"For example, 'a LAN covers a small geographical area' on its own is a true fact but does not answer the question.

However, 'a LAN covers a smaller geographical area than a WAN' answers the question and would gain credit"

(Examiner report, March 2022, Paper 9626/32, p. 6). *"whilst candidates do have to show some awareness that they are comparing, there is no requirement for candidates to present both sides of the argument"*

(Examiner report, June 2025, Paper 9626/12, p. 6).

2. Write each difference as one sentence about both things, joined by 'whereas'. Do not write one paragraph about A and then another paragraph about B. Examiners gave no marks when they could not see the comparison.

"differences are best presented together, perhaps joined by the use of 'whereas'. In some cases, candidates wrote two separate descriptions one of phishing and the other of vishing" (Examiner report, November 2022, Paper 9626/12, p. 3).

"this clarity was sometimes lacking and so marks were not awarded"

(Examiner report, November 2022, Paper 9626/12, p. 3).

3. Do not answer with a bullet list or a table. Write full sentences. One examiner report says lists of points, bulleted or in tables, are unlikely to get full marks for a comparison.

"candidates who provide lists of points, bulleted or in tables, are unlikely to gain full marks because the question asks for comparisons which are best written in free response" (Examiner report, June 2023, Paper 9626/32, p. 21). *"Weaker answers just stated facts, sometimes as unconnected bullets, or described uses of LANs/WANs"*

(Examiner report, March 2022, Paper 9626/32, p. 6).

Answer the question asked

1. Do not repeat the question as a similarity. If the question says both devices exchange data, 'both exchange data' gets no mark. The examiners also called very general similarities weak, such as 'both are plastic cards'.

"answers that repeat the question, in this case stating that both exchange data in networks, will not be given credit"

(Examiner report, June 2023, Paper 9626/32, p. 21). *"Weaker answers referred to both being plastic cards"*

(Examiner report, June 2023, Paper 9626/31, p. 17).

2. Compare exactly the two things named in the question. Write about both of them. Do not answer about something else that just shares a keyword.

"some candidates either only focussed on one of the two methods, or did not discuss the impact of each method on the quality of a video file" (Examiner report, November 2025, Paper 9626/12, p. 7). *"a significant number of responses"*

referred to dynamic IP assignment using DHCP. Such references did not answer the question" (Examiner report, March 2023, Paper 9626/32, p. 8).

Worked example

Q3 [6 marks] "Compare the use of static and dynamic routing when sending data packets through IP networks. You must include similarities and differences." Question paper 9626/32, March 2023, p. 4

A full-mark answer, built from the mark scheme (not a real candidate's):

Both static and dynamic routing are used in routers to decide the next hop that a data packet is sent to.¹ Both types of routing provide a default route for a packet if no other route can be found.² Network administrators create static routing tables, whereas algorithms create dynamic routing tables.³ Static routes are fixed, whereas dynamic routing lets packets take different routes depending on network conditions.⁴ Static routing tables stay the same when network conditions or devices change, whereas dynamic routing tables update to match the changes.⁵ Static route information does not time out, whereas dynamic routes have a time-to-live.⁶

Where the marks come from ¹ "Used in routers/routing devices to determine next node/router/device/hop to send data packet/datagram/frame". ² "Both provide default route for packet if no route can be determined". ³ "Static routing tables created by network administrators whereas dynamic routing tables are created by algorithms". ⁴ "Static routes are fixed whereas dynamic routing allows packets to take different routes depending on network conditions". ⁵ "Static routing tables not updated/amended when network conditions/device change whereas dynamic routing are updated/can adjust to changing network conditions". ⁶ "Static routing table/info does not 'time out' whereas dynamic routes can be time-limited/have time-to-live data". (Mark scheme 9626/32, March 2023, p. 5)

A weak answer of the kind the examiners describe (written for this guide, about 0 of 6):

Static IP addresses are set by the network administrator and do not change. Dynamic IP addresses are given out by a DHCP server and can change each time a device connects.

It answers about IP addresses, but the question is about routing. It has no similarity. It also describes each type on its own, with no comparison.

"It was noted that a significant number of responses referred to dynamic IP assignment using DHCP. Such references did not answer the question and it can only be assumed that the candidates spotted the words 'dynamic' and 'IP' and misunderstood the question" (Examiner report, March 2023, Paper 9626/32, p. 8).

Worked example

Q1 [6 marks] "Contrast the characteristics of local area networks (LANs) and wide area networks (WANs)." Question paper 9626/32, March 2022, p. 2

A full-mark answer, built from the mark scheme (not a real candidate's):

A LAN covers a smaller geographical area than a WAN.¹ A LAN has a higher data transfer speed than a WAN.² One organisation usually owns a LAN, whereas a number of organisations often own a WAN.³ LANs have fewer data transmission errors than WANs.⁴ LANs have fewer network congestion problems than WANs.⁵ A LAN uses Layer 1 or 2 devices such as switches and hubs, whereas a WAN uses Layer 3 devices such as routers.⁶

Where the marks come from ¹ "LAN covers a smaller geographical area/valid examples than WAN". ² "LAN has higher data transfer speed/greater bandwidth than WAN". ³ "LANs are (usually) owned by one organisation whereas WANs are (usually/often) owned by a number/collective ownership". ⁴ "LANs have fewer data transmission errors than WANs". ⁵ "LANs have fewer network congestion problems than WANs". ⁶ "LANs use Layer 1 or 2 devices/switches/bridges/hubs/repeaters whereas WANs use Layer 3 devices/Routers/Multi-layer Switches".
(Mark scheme 9626/32, March 2022, p. 3)

A weak answer of the kind the examiners describe (written for this guide, about 0 of 6):

- LAN: small area, one building.
- WAN: large area, the internet.
- Both are networks that connect computers.

Each bullet states facts about one network, so no bullet gives a difference. The last bullet is a similarity, and Contrast gives no credit for similarities.

"Weaker answers just stated facts, sometimes as unconnected bullets, or described uses of LANs/WANs. Others gave similarities. These did not gain credit because the responses did not answer the question"

(Examiner report, March 2022, Paper 9626/32, p. 6).

Identify, State, Give, Define, Name, Suggest

Identify. Cambridge's definition: "name/select/recognise" Syllabus 2025 to 2027, p. 66

State. Cambridge's definition: "express in clear terms" Syllabus 2025 to 2027, p. 66

Define. Cambridge's definition: "give precise meaning" Syllabus 2025 to 2027, p. 66

Suggest. Cambridge's definition: "apply knowledge and understanding to situations where there are a range of valid responses in order to make proposals/put forward considerations" Syllabus 2025 to 2027, p. 66

Give: not in the syllabus list of command words. Not in the syllabus command-word table (page 66).

Name: not in the syllabus list of command words. Not in the syllabus command-word table (page 66).

List: not in the syllabus list of command words. Not in the syllabus command-word table (page 66).

How much to write

1. For Identify, State and Name, a single word or short term can be enough. Examiners described one Identify question as looking for four one-word answers. Another question wanted only the name or acronym of a protocol.

"In effect, this question was a test of memory recall, as whilst the answer did not require precise wording, it was looking for four one-word answers" (Examiner report, June 2024, Paper 9626/11, p. 2). *"This question required only a name or acronym of a protocol that works at the internet layer"* (Examiner report, June 2025, Paper 9626/31, p. 19).

2. When the question asks you to describe, write more than one word for each answer.

"As candidates were asked to describe, rather than state, answers needed to be more than one-word answers" (Examiner report, November 2025, Paper 9626/11, p. 3).

3. Short-answer questions usually give one mark for each correct item. The mark scheme says 'One from' or 'Four from'.

"ONE from: • a cryptographic hash of the previous block • a timestamp • (all the) transaction data"

4. Some questions say 'identify and describe', or ask for an item and a description. The mark scheme can then use matched pairs. You get one mark for the item and one mark for a description that goes with it.
"Max two for each matched pair" (Mark scheme 9626/12, March 2024, p. 4). *"Max FOUR marks, TWO per paired answer"* (Mark scheme 9626/12, June 2025, p. 10).
5. Give exactly the number of answers that the question asks for. Examiners mark only the first answers, up to that number. Extra answers earn nothing.
"Where candidates are asked for a specific number of answers, only the first of these up to that number will be marked. Therefore, for this question, if a candidate gave more than four answers, only the first four were marked" (Examiner report, November 2023, Paper 9626/13, p. 8).

Use the exact term

1. Make a definition precise. Examiners praised short, exact definitions such as 'a sequence of characters'. They called nearly-right answers a common mistake. One example is saying that encrypted data is unreadable. In fact the data is readable, but its meaning is hidden. Learn the exact meaning of each syllabus term.
"Many candidates could give the precise definition of a string as a 'sequence of characters'" (Examiner report, November 2024, Paper 9626/33, p. 18). *"One common mistake is considering that encrypted data is unreadable without a key, which is one advantage of encrypting the data"* (Examiner report, June 2023, Paper 9626/11, p. 2).
2. Use the exact names from the syllabus. Made-up names such as 'updating maintenance' got no marks. Be as specific as the question asks. For example, 'numeric' was not enough when the question asked which numeric data type.
"Poorer answers incorrectly named the maintenance as e.g., 'updating maintenance' or 'replacement' maintenance. The names of the types of maintenance are given in Topic 16.8 so candidates are reminded to use these" (Examiner report, November 2023, Paper 9626/32, p. 18). *"Typical issues included stating that the field was numeric, but not stating what type of numeric field was best, or simply confusing the data types"* (Examiner report, November 2024, Paper 9626/12, p. 4).

Answer this question

1. Do not give an item from the question as your answer. If the question already names an item or gives an example, repeating it earns nothing. A new wording of the question also earns nothing, for example 'a command line interface uses commands'.
"Candidates should not repeat the question in their answers. For example, Question 5(a) asks for one 'other item of data' but a significant number of candidates included the items given in the question stem in their answer, which did not gain them credit" (Examiner report, June 2025, Paper 9626/33, p. 24). *"For example, candidates stated that a command line interface used commands or a gesture-based interface used gestures. In cases where candidates answered in these terms, without any further expansion, no credit could be awarded"* (Examiner report, November 2023, Paper 9626/12, p. 6).
2. Answer the actual question. If it asks you to give reasons, a description gets no marks. If it asks how something could go wrong, a definition gets no marks.
"Many candidates described tunnelling which did not answer the question. The question required candidates to give reasons why it would be used" (Examiner report, November 2022, Paper 9626/32, p. 16). *"Candidates who failed to achieve marks usually did so because they gave a definition of visual checking, rather than answer the question"* (Examiner report, March 2025, Paper 9626/12, p. 2).
3. Fit short answers to the scenario too. Examiners did not accept answers about the wrong part of the scenario.
"Where candidates did not achieve full marks, this was often because they listed issues to do with the train service"

itself" (Examiner report, November 2025, Paper 9626/12, p. 5). "When faced with a question that has a clear context, learners should take this as an opportunity to apply their understanding in the particular context" (Examiner report, June 2023, Paper 9626/12, p. 6).

Suggest needs reasons that fit

1. Suggest is rare, with 2 parts from 2022 onward. One of them has an examiner comment, on improving a flowchart. Each suggestion had to be relevant and had to fit clearly on the flowchart. A 2021 report on a Suggest question criticised answers that said what something was. Those answers gave no reasons that fitted the scenario.

"As the question asks for improvement to the flowchart, candidates had to ensure that suggested improvements were both relevant and able to be clearly placed on the flowchart" (Examiner report, November 2023, Paper 9626/13, p. 10).
"While many responses provided some detail about what incremental prototyping involved, most did not give reasons why it might be used when developing a spreadsheet for managing finances" (Examiner report, June 2021, Paper 9626/32, p. 20).

Worked example

Q2 [4 marks] "Identify four factors which can affect the quality of information."

Question paper 9626/11, June 2024, p. 2

A full-mark answer, built from the mark scheme (not a real candidate's):

1 Accuracy¹ 2 Relevance² 3 Age³ 4 Completeness⁴

Where the marks come from ¹"Accuracy (1)". ²"Relevance (1)". ³"Age (1)". ⁴"Completeness (1)".
(Mark scheme 9626/11, June 2024, p. 3)

The mark scheme also accepts Level of detail.

Worked example

Q5(a) [1 mark] "Blockchains contain the records and the cryptographic hashes used to secure the records. State one other item of data that makes up the blockchain." Question paper 9626/33, June 2025, p. 6

A full-mark answer, built from the mark scheme (not a real candidate's):

A timestamp.¹

Where the marks come from ¹"a timestamp". (Mark scheme 9626/33, June 2025, p. 7)

A weak answer of the kind the examiners describe (written for this guide, about 0 of 1):

The cryptographic hashes that secure the records.

The question already gives this item. Repeating the question stem earns nothing.

"a significant number of candidates included the items given in the question stem in their answer, which did not gain them credit" (Examiner report, June 2025, Paper 9626/33, p. 24).

'Other' in the question means an item that the stem does not already give.

Worked example

Q1(a) [1 mark] "JavaScript can manipulate and store strings in variables. (a) Define the term string as used in JavaScript." Question paper 9626/33, November 2024, p. 2

A full-mark answer, built from the mark scheme (not a real candidate's):

A string is a sequence of characters.¹

Where the marks come from ¹ "(An object representing) a sequence of (any) characters".
(Mark scheme 9626/33, November 2024, p. 5)

Complete, Draw, Write (algorithms and flowcharts)

Write (an algorithm / pseudocode): not in the syllabus table of command words. Not in the command-word table (page 66). This is the wording of syllabus topic 4.1. The ellipses are part of the syllabus keywords.

"Writing an algorithm using pseudocode to solve a given problem, including: – INPUT/READ – WRITE/PRINT – FOR... NEXT – FOR...NEXT...STEP – IF...ELSE...ENDIF – WHILE...ENDWHILE – REPEAT...UNTIL – CASE...ENDCASE"
(Syllabus 2025 to 2027, p. 21).

Complete (an algorithm or flowchart): not in the syllabus table of command words. Not in the command-word table (page 66). Topic 4.1 wording.

"Writing and editing a given algorithm that demonstrates a decision-making process, including: – conditional branching – looping – nested loops – procedures/subroutines" (Syllabus 2025 to 2027, p. 21).

Draw (a flowchart): not in the syllabus table of command words. Not in the command-word table (page 66). This is the syllabus symbol list on page 61 that examiners expect you to use.

"Flowchart symbols Element Symbol Input/output Decision Terminator (Start/Stop) Process box Subroutine Connector A Flow line" (Syllabus 2025 to 2027, p. 61).

One mark per feature

1. Examiners give one mark for each feature the answer needs, such as an IF...ENDIF or outputs that match the conditions. Most of these marks stand alone, and a partly wrong answer still scores. A few marks depend on other marks. Always attempt the question. The examiner reports say the easier marks are generally accessible.

"Use of one IF...ENDIF (1) • All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded) (1)"
(Mark scheme 9626/13, June 2025, p. 7). "While questions about flowcharts and pseudocode often test a range of skills, easier marks are generally accessible, making it disappointing to see some candidates choose not to attempt this question" (Examiner report, June 2024, Paper 9626/13, p. 11).

Write pseudocode the examiners accept

1. Use the structures and keywords from the syllabus (IF...ELSE...ENDIF, WHILE...ENDWHILE, REPEAT...UNTIL, CASE...ENDCASE). If the question names a structure, use it. Students who mixed CASE with IF statements lost marks.

"some candidates did not use the coding structures and terms given for this qualification"
(Examiner report, November 2023, Paper 9626/11, p. 1). "Most candidates tried to combine the structure with IF

statements, thereby highlighting this lack of understanding" (Examiner report, June 2024, Paper 9626/11, p. 3).

2. Write the answer as pseudocode. Students who wrote prose lost marks. Cambridge publishes a pseudocode guide that shows the expected layout.

"proved challenging to candidates who wrote their answer in prose rather than using algorithmic constructs"

(Examiner report, June 2022, Paper 9626/11, p. 2). *"Centres are reminded that CIE has produced a pseudocode guide for teachers and candidates working on this course"* (Examiner report, June 2025, Paper 9626/13, p. 11).

3. Initialise counters and totals before the loop starts. In June 2025 Paper 13 Q7, initialisation inside the loop was the most common mistake among otherwise strong candidates. In Q13 of the same paper, a missing initialisation was the usual reason for losing marks.

"a sizeable minority placed the initialisation of the count within the loop"

(Examiner report, June 2025, Paper 9626/13, p. 11). *"Where candidates did not achieve full marks, this was generally because they did not initialise the variable Sum"* (Examiner report, June 2025, Paper 9626/13, p. 12).

4. Put the counter increment inside the loop. An increment that sits only inside an IF within the loop does not get the mark. Make the loop stop at the right value.

"Correct Increment of Count (1) • Increment of Count is within the loop (1) • UNTIL Count = 20 (1) • UNTIL in correct position (1)" (Mark scheme 9626/13, June 2025, p. 7). *"count ← count + 1 (if within while statement but not if within IF statement (unless in both))"* (Mark scheme 9626/12, November 2023, p. 8).

5. Small syntax errors cost marks. Write each comparison in full (IF X > Y AND X < Z, not IF X > Y and < Z). Close every IF with ENDIF. Input each variable separately ('input X and Y' does not work). Put text in speech marks, with commas between text and variables.

"treating the test criteria as a string and using the structure IF X>Y and <Z, instead of IF X>Y AND X<Z"

(Examiner report, June 2025, Paper 9626/13, p. 11). *"Using an input statement 'input X and Y'. This would not correctly input the values for the two variables. • Not indenting within the IF statement and not closing the IF statement. • Not using speech marks and commas within the outputs"* (Examiner report, November 2023, Paper 9626/13, p. 10).

6. Indent the lines inside IF, ELSE and loops. Some mark schemes give a mark just for correct indentation.

"IF and ENDIF indented correctly (1)" (Mark scheme 9626/13, November 2023, p. 8). *"Where marks were awarded, these tended to be for structural awareness, such as correct use of indentation"*

(Examiner report, June 2024, Paper 9626/11, p. 4).

7. Write variable names of two or more words with no spaces. Examiners do not take off marks for a slightly wrong variable name if it is clear which variable you mean.

"It should be noted that where variable names are made up of two or more words, there should be no gap between these words" (Examiner report, November 2022, Paper 9626/12, p. 5).

"(Ignore errors with variable names, as long as it is clear which variable is being referred to)" (Mark scheme 9626/12, November 2022, p. 8).

Draw flowcharts with the syllabus symbols

1. Use the symbols shown in the syllabus: parallelograms for input and output, rectangles for processes, and terminators for Start and Stop. Do not use other versions of the symbols.

"While it is accepted that there are variations on the use of symbols and their naming in system flowcharts, it is expected that for the purposes of this syllabus that the symbols shown in the syllabus be used"

(Examiner report, November 2024, Paper 9626/33, p. 20). *"Candidates need to use the symbols provided by the board and should have practiced using them correctly"* (Examiner report, March 2025, Paper 9626/12, p. 4).

2. Do not mix up input and process boxes. Mark schemes give marks for correct shapes and for arrowheads on flow lines.

"In a small number of cases, candidates confused input symbols with process symbols"

(Examiner report, November 2024, Paper 9626/12, p. 4). *"All lines have arrow heads with direction of flow 1 mark 9 All*

3. Give every decision box two exits labelled Yes and No (True and False are accepted). Write the condition as a maths expression. The examiners did not accept words such as 'is x between 5 and 10'.

"Candidates are reminded that all decision boxes must have a Yes and a No output (True and False are acceptable) and that they should use mathematical expression rather than text in their boxes. For example, the term 'is x between 5 and 10' is not acceptable" (Examiner report, June 2023, Paper 9626/11, p. 4). *"each decision has two outcomes. These outcomes may be yes/no, or true/false, but they must be shown and labelled"* (Examiner report, March 2025, Paper 9626/12, p. 4).

Do what the question asks

1. Do exactly what the question asks. If it says 'redraw', draw a new flowchart and leave the printed one alone. If it asks for a flowchart, do not write pseudocode.

"candidates were clearly asked to redraw the flowchart. Candidates who amended the original flowchart and did not provide a new flowchart were therefore not answering the question" (Examiner report, June 2025, Paper 9626/13, p. 12). *"In a very small number of cases, candidates gave a form of pseudocode, instead of a flowchart for the answer"* (Examiner report, November 2024, Paper 9626/12, p. 4).

2. Include every feature the question lists, such as a calculation and an output.

"As a general comment on questions such as these, most candidates did not include all features identified in the question" (Examiner report, November 2023, Paper 9626/11, p. 3).

3. When you suggest or describe a change to a flowchart or pseudocode, say exactly where it goes. For example, say which line you remove and what replaces it. Write the answer in the answer space, not on the printed code.

"candidates need to be extremely clear about which line is being removed, and how it is to be replaced" (Examiner report, June 2025, Paper 9626/11, p. 3). *"centres are advised that candidates should answer the question in the space provided in the question paper, rather than writing answers on the pseudocode included in the question paper"* (Examiner report, November 2025, Paper 9626/12, p. 6).

Worked example

Q7 [8 marks] "Write a pseudocode algorithm using REPEAT...UNTIL which allows the teacher to input 20 test marks and outputs the correct message for each inputted mark."

Question paper 9626/13, June 2025, p. 6

Marks above 70 get a Distinction. Marks from 55 to 70 get a Merit, and marks from 40 to 54 get a Pass. Marks below 40 get the message 'You have not reached the Pass mark'.

A full-mark answer, built from the mark scheme (not a real candidate's):

```
Count ← 0 1
REPEAT 2
    INPUT Mark 3
    IF Mark > 70 THEN
        PRINT "You have achieved a Distinction" 4
    ELSE
        IF Mark > 54 THEN
            PRINT "You have achieved a Merit" 5
        ELSE
            IF Mark > 39 THEN
                PRINT "You have achieved a Pass"
```

```

        ELSE
            PRINT "You have not reached the Pass mark" 6
        ENDIF
    ENDIF
ENDIF 7
Count ← Count + 1 8
UNTIL Count = 20 9

```

Where the marks come from 1 "Initialise Count before Mark variable is input (1)". 2 "REPEAT in correct logical location (1)". 3 "Input Mark (1)". 4 "Correct Distinction grade criterion/correct fail criterion (1)". 5 "Correct Merit grade criterion/correct Pass grade criterion (1)". 6 "All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded) (1)". 7 "Use of one IF...ENDIF (1)". 8 "Correct Increment of Count (1) • Increment of Count is within the loop (1)". 9 "UNTIL Count = 20 (1) • UNTIL in correct position (1)". (Mark scheme 9626/13, June 2025, p. 7)

A weak answer of the kind the examiners describe (written for this guide, about 5 of 8):

```

REPEAT Count ← 0 INPUT Mark IF Mark >= 55 and <= 70 THEN OUTPUT "You have achieved a Merit" ... Count ← Count + 1 UNTIL Count = 20

```

The code sets Count to 0 on every pass. The loop never ends, and the answer loses the initialisation mark. The Merit test does not repeat Mark before <= 70, and the IF has no ENDIF. The answer still gets the REPEAT, INPUT, increment and UNTIL marks.

"However, a sizeable minority placed the initialisation of the count within the loop. This was the most frequently made mistakes amongst candidates who otherwise had a strong understanding of what was required"
(Examiner report, June 2025, Paper 9626/13, p. 11).

The syllabus lists only the operators >, < and =, so this answer nests IF...ELSE in place of AND or >=.

Worked example

Q4 [7 marks] "Draw a program flowchart to represent this algorithm: • Input an integer value. • Repeat this until you have entered 20 values. • At the end, output the largest value and the average of the values." Question paper 9626/12, November 2024, p. 5

A full-mark answer, built from the mark scheme (not a real candidate's):

Start with a Start terminator and finish with an End (or Stop) terminator. 1
In a process rectangle before the loop, set COUNT and TOTAL to 0. 2
Use parallelograms for every input of NUM and for the output of MAX and the average. 3
Input the first NUM and set MAX ← NUM. Then, inside the loop, use process rectangles for TOTAL ← TOTAL + NUM and COUNT ← COUNT + 1. 4
Draw two decision diamonds, 'NUM > MAX?' and 'COUNT = 20?', each with labelled Yes and No exits. 5
From 'NUM > MAX?', Yes goes to MAX ← NUM and No skips it. From 'COUNT = 20?', No

goes to an input box for the next NUM and back into the loop, and Yes leaves the loop. ⁶

After the loop, calculate $AVERAGE \leftarrow TOTAL / 20$ in a process rectangle. ⁷

Where the marks come from ¹"Start and End (accept STOP) terminators (1)". ²"Two variables (e.g. MAX, COUNT, TOTAL, NUM) initialised/set to zero before loop (1)". ³"Input box and output box are both parallelograms (1)". ⁴"Sum all values input (1) 11 Set new MAX when input value > MAX in loop 12 Increment COUNT (1)". ⁵"TWO Correctly drawn decision boxes with YES/NO decisions (1)". ⁶"All YES/NO decisions (used) go to correct outcomes (i.e. all logic is correct) (1)". ⁷"Correctly calculate average (must include process/calculation) (1)".

(Mark scheme 9626/12, November 2024, p. 5)

A weak answer of the kind the examiners describe (written for this guide, about 2 of 7):

A flowchart that puts 'Input number' in a rectangle, has a decision box 'Have 20 numbers been entered' with no Yes or No labels, and never sets COUNT or TOTAL to zero.

The input sits in a rectangle, which is the wrong symbol. The decision is in words and has no Yes or No labels. COUNT and TOTAL never start at zero.

"In a small number of cases, candidates confused input symbols with process symbols, but overall, performance on this question was very good" (Examiner report, November 2024, Paper 9626/12, p. 4).

Setting MAX to the first value makes the flowchart work when every number is negative, and the mark scheme also accepts MAX set to 0.

Sources

Each point quotes the Cambridge document it comes from, with the page number. There are 384 quotations from 70 documents. They are the 9626 syllabus for 2025 to 2027, the Cambridge Learner Guide for 9626, and the Paper 1 and Paper 3 question papers, mark schemes and examiner reports from 2017 to March 2026. A script compared every quotation with its source, word for word, before this PDF was made.

The worked examples use real questions. The full-mark answers join mark scheme points into sentences, and the numbers show which point each sentence earns. The weak answers show a mistake that an examiner report describes. No answer in this booklet was written by a real candidate.

Cambridge also publishes Example Candidate Responses for 9626, with real scripts and examiner comments. They are on the School Support Hub, which needs a teacher login, so they are not quoted here.

The counts of question parts come from Papers 1 and 3, 2022 to March 2026, read by a script and checked by hand where the script was unsure. A part with no command word of its own, such as '(b) Trojan', is counted as having none. Papers 2 and 4 are practical papers and are not included.

How much evidence there is for each word

Describe	409 Describe parts on Papers 1 and 3 from 2022 to March 2026, 388 of them with a mark scheme. 83 of those 388 mark schemes use the (1st) label. Cambridge's mark-scheme key does not define (1st).
Explain	176 Explain parts on Papers 1 and 3 from 2022 to March 2026, 174 of them with a mark scheme. 38 of those 174 mark schemes use the (1st) and (1) labels. The rest give one mark for each point. The limit for answers that are 'all only first marks' appears in one mark scheme, November 2023 mark scheme (Paper 13) Q3(c).
Discuss	Not thin. Discuss appears in 80 question parts from 2022 onward, and 77 of them have mark schemes. We made the counts above ourselves from the mark-scheme text. They could be off by one or two.
Evaluate (and Assess)	Evaluate has plenty of evidence: 25 question parts from 2022 onward (114 in all years). Assess is in the syllabus list of command words. But no question part uses Assess as its command word. A search of all question paper pages found no question that starts with 'Assess'. There are no Assess mark schemes or examiner report comments. So this report gives no separate advice for Assess beyond its syllabus definition ('make an informed judgement'). The count of conclusion marks comes from a text search of the mark schemes for 'conclusion' or 'overall judgement'.
Analyse	There are 19 Analyse parts from 2022 onward (18 with a matched mark scheme), mostly on Paper 3. That is enough to see the pattern, but fewer than for Evaluate or Discuss. One examiner report comment (June 2024, Paper 32, Q8) says you could get full marks without both sides. So 'cover both sides' is advice, not a fixed rule.

Justify	There are 20 Justify parts from 2022 onward. One of them (March 2026 Q10(c)) has no matched mark scheme. Many examiner reports from 2022 to 2025 repeat the one-sided rule. The 'drawbacks of alternatives' exception rests on little evidence. It comes from one examiner report sentence (June 2024, repeated for Papers 31 and 33), one 2023 examiner report comment and one mark scheme. So use it carefully.
Compare and Contrast	Compare: 16 question parts from 2022 to 2025, all 16 with mark schemes. Contrast: 6 parts from 2022 to 2026, 5 with mark schemes. The sixth part, March 2026 Paper 12 Q7(b), has no mark scheme in the source set. November 2024 Paper 33 Q3(b) is left out because 'contrast' there is an image term in a Describe question. The examiner reports disagree on whether Compare needs both sides when the question does not ask for both.
Identify, State, Give, Define, Name, Suggest	Papers 1 and 3 from 2022 to March 2026. Parts: Identify 28, State 12, Give 11, Define 8, List 2, Suggest 2, Name 1. Parts with a mark scheme: Identify 26, State 8, Give 11, Define 6, List 2, Suggest 2, Name 0. The two List parts also ask you to describe. The one Name part (March 2026, Paper 12 Q10) says 'identify' in its wording. One Suggest question from 2022 onward has an examiner comment (November 2023 question paper (Paper 13) Q8). The Suggest advice also uses one 2021 report. Give, Name and List are not in the syllabus command-word table. No examiner report says whether an example in a definition gains or loses marks.
Complete, Draw, Write (algorithms and flowcharts)	Write: 7 question parts from 2022 to 2025, all 7 with mark schemes. Complete: 17 parts from 2022 to 2026, 15 with mark schemes. Draw: 9 parts from 2022 to 2025, 8 with mark schemes. Most algorithm and program-flowchart questions are in Paper 1. In Paper 3, Draw and Complete also tag DFD, PERT and system-flowchart parts, and only the symbol points apply to those. Write also tags November 2025 Paper 11 and Paper 13 Q8(b), which are about audio. The June 2022 examiner report says several times that algorithms and flowcharts were a recent addition to the syllabus.